

Dynamics on aggregated connectomes in the browser: normalization, structural bias, and honest tuning lessons across two connectomes

Technical note v4.1, 2026-09-27. Code: [fruitfly/game/](#) — v1 measurements @ commit [d92807b](#), v2 probes @ [73da541](#), v3–v4 scripts & experiments @ [63c54ca](#) / [709b188](#), this revision @ [80a4768](#) (MIT). Data: FlyWire FAFB v783 (CC BY-NC-SA 4.0) and MaleCNS v1.0 (CC BY 4.0). v4.1 closes the third review round ("minor revisions, no new experiments required"): histamine-account correction, reverse-experiment precision, two final frozen reproduction entries, and editorial cleanup. Point-by-point responses: [docs/REVIEW-RESPONSE.md](#).

Repository: <https://github.com/professorwang/flybrain-banana-quest> · Demo: <https://professorwang.github.io/flybrain-banana-quest/> (?dataset=malecns 切换 MaleCNS)

中文摘要

本文记录浏览器游戏《电子果蝇·香蕉大作战》(Banana Quest) 在真实连接组上跑动力学时实测到的方法学问题及其跨数据集复测结果，供连接组仿真社区参考。游戏用 LIF 脉冲网络在两个真实连接组 (FlyWire FAFB v783: 139,255 神经元 / 2,698,236 条聚合连接; MaleCNS v1.0: 176,422 神经元 / 6,287,749 条 ≥ 5 突触连接) 上驱动一只 2D 果蝇觅食——连接组放电参与运动读出与进食门控，同时游戏包含显式运动规则 (§1.1 如实列出)。发现一：参考实现沿用的全局 max 权重归一化 (增益 0.15) 在**参考增益下传播失败**——重尾突触计数分布把典型权重压到阈值的约 1/2000，下游膜电压在 100 ticks 内观察到的最大值仍不过阈 (FlyWire 0.780、MaleCNS 0.977)，中枢放电为零；增益放大 10 倍后两数据集分别有 16 / 131 次中枢放电 (评审复测，我们一票复现)，因此结论收窄为“参考增益下失败”而非“必然失败”。可行替代是本项目采用的 **postsynaptic L1 归一化** (每个突触后神经元总入权重归一化为常数；与 Shiu et al. 2024 的每突触固定系数方案不同，早期版本的误引已更正)，工作点为数据集相关 (FlyWire 3.0 / MaleCNS 4.0)。发现二：FlyWire 下行池按胞体 x 切半后存在结构性偏置 (任一侧刺激右池都更活跃，有效差仅约 1.3%)，转向读出采用本项目选用的启发式高通读出 (慢速 EMA 基线的偏差)；该偏置在所测 MaleCNS 条件下未复现 (侧向对比方向正确，差 9.49 个百分点，算式 $91/157 - 111/229 \approx 0.09490$)，“分组伪影”解释目前仅为待验证假说，决定性检验是在同一 FlyWire 数据上重分组。发现三 (方法论)：小种子扫描会过拟合——实测冠军配置在 10 个样本外种子上未保持优势 (且首吃时间只统计成功 episode)；用于选关卡的种子集不再独立。符号实验闭环：仅把 MaleCNS 的谷氨酸符号从抑制翻为兴奋 (其余不变)， $t_i=4$ 时视叶出现高活动 (VIS_OL 553,863 次放电)；反向把 FlyWire 的谷氨酸翻为抑制， $t_i=4$ 视叶高活动消失 (VIS_ME 405,788 $\rightarrow$ 0)——在 $t_i=4$ 、 $I=1$ 、200 tick 及所比较的 **GLUT 正/负两种配置下**，谷氨酸符号赋值既是高视叶活动表型的充分条件也是必要条件 ($t_i=3$ 结果照实报告但不纳入该结论范围)；两数据包符号表不同 (GLUT 兴奋 vs 抑制；MaleCNS 明确将 8,021 个 histamine 标记节点设为抑制，而 FlyWire 的六类递质预测不含 histamine——不能从缺少该预测类别直接推断真实组胺能神经元在 FlyWire 包中的赋号)，是跨数据组动力学比较必须显式声明的混杂。所有数字均可由文中命令复现 (注明历史记录者除外)。

Abstract

Banana Quest is a zero-dependency browser game in which a leaky integrate-and-fire (LIF) network running on a real connectome helps drive a 2D foraging fly. Connectome activity participates in the motor readout and feeding gating; the game additionally contains explicit motor rules, which we list openly (§1.1). We report three methodological findings and their cross-dataset replication on FlyWire FAFB v783 and MaleCNS v1.0. (1) Global-max weight normalization **at the reference gain (0.15)** fails to propagate activity on aggregated connectomes with heavy-tailed synapse-count distributions — the maximum membrane voltages observed within 100 ticks stay below threshold (0.780 / 0.977), with zero central spikes on both datasets; at 10× gain a small number of central spikes appears (16 / 131), with unknown stability, so we scope the claim to the reference gain rather than "global-max never propagates". A working alternative is postsynaptic L1 normalization (adopted in this project; distinct from Shiu et al.'s fixed per-synapse coefficient scheme), with a dataset-dependent working point (3.0 / 4.0). (2) A coarse bilateral descending readout on FlyWire carried a structural bias larger than the stimulus-driven signal (~1.3%); it did not replicate under the tested MaleCNS conditions (correctly signed contrast, 9.49 percentage points, $91/157 - 111/229 \approx 0.09490$), so the "pool-definition artifact" explanation remains a hypothesis — the decisive test is re-pooling on the same dataset. (3) Small-seed sweeps of high-variance behavioral metrics can overfit: concretely, our 5-seed champion configuration did not retain its advantage on 10 out-of-sample seeds (first-eat statistics count successful episodes only). Sign experiments now close both directions: flipping only the glutamate sign from inhibitory to excitatory in MaleCNS produces high visual-lobe activity at $t_i=4$ (553,863 VIS_OL spikes), and the reverse flip in FlyWire abolishes it (VIS_ME 405,788 $\rightarrow$ 0) — **at $t_i=4$, $l=1$, 200 ticks, and across the two GLUT sign configurations compared**, the glutamate sign assignment is both sufficient and necessary for the high-activity phenotype ($t_i=3$ results are reported but excluded from this conclusion). The two packages also differ in their sign tables: GLUT excitatory vs inhibitory, and MaleCNS explicitly marks 8,021 histamine-labeled nodes inhibitory while FlyWire's six-class transmitter prediction contains no histamine class at all — the absence of a predicted class cannot be used to infer how real histaminergic neurons are signed in the FlyWire package. These are confounds any cross-package dynamical comparison must declare. All numbers are reproducible with the commands given, except where marked as historical records.

1. Background

Banana Quest (电子果蝇 香蕉大作战) is an open-source, zero-build browser game. A virtual fly forages for bananas in a 2D arena. Olfactory input is injected as current into left/right pools of olfactory receptor neurons; spikes propagate through the connectome to descending neurons, whose pooled left/right firing-rate difference steers the fly, and whose total rate, scaled by a hunger variable, sets its speed. Feeding is gated by gustatory stimulation driving a proboscis motor-neuron readout. **Connectome spiking participates in steering and in feeding gating — but it is not the only thing moving the fly: the game also contains explicit motor rules, disclosed in §1.1.**

The FlyWire connectome ships as a compact binary (12.4 MB gzipped) packaged by the community project `snede/flybrain` (MIT): FAFB v783 edges aggregated per neuron pair with signed weights, plus a coarse 63-group functional annotation. Their reference simulator runs the full network in a Web Worker at ~10 ticks/s using a neuropil-gated dormant-group optimization; our simulation core (`src/sim-core.js`) is adapted from it and runs unchanged in both the browser worker and Node.js for testing. The second dataset is MaleCNS v1.0 (neuPrint `male-cns:v1.0`; 176,422 neurons, 6,287,749 connections at ≥ 5 synapses; the full male central nervous system including the VNC; CC BY 4.0), selected at runtime with `?dataset=malecns`. We take the FLYB v1 binary bundled by the fly-brain-minecraft project (MIT) and convert it with our own `tools/flyb_to_bin.py` into the same `sim-core` format — 26 functional groups with the same region encoding, and pools filtered on MaleCNS's native annotations (`somaSide`, `superclass`, `type`). Every probe below runs through the identical `sim-core.js` on both datasets.

1.1 Explicit game-layer motor rules (disclosed in v3)

Earlier versions of this note said "no hand-crafted behavior layer" — that was overstated. The brain participates in behavior, but the following explicit rules also move or stop the fly, and reviewers rightly asked for them to be listed:

- **Standby random walk:** when the descending pools' total firing rate falls below a threshold, the fly performs a slow random walk driven by game-layer noise, explicitly labeled "待机噪声 (非脑驱动)" in the UI.

- **Approach deceleration and stopping:** speed is additionally scaled down when turning sharply and when close to the banana (a turn-and-slow kinematic rule), and the fly stops inside the eating radius while the feeding gate is evaluated.
- **Scripted escape maneuver:** after a startle, the escape trigger is the brain's response (descending-rate surge or DNp01 firing), but the escape maneuver itself — a random sharp turn plus a fixed-duration speed boost — is scripted.
- **Wall bounce:** heading reflects off arena walls with added noise.
- **Hunger wiring:** hunger is a scripted clock that multiplies olfactory gain and injects current into a hunger-drive group (an acknowledged artificial wiring).
- **Sensor mapping and pool selection:** the odor→current mapping, the left/right pool splits, and all readout pools are hand-designed (acknowledged throughout; MaleCNS stimulus pools are additionally asymmetric, §5.5).

What the brain does do: it gates eating (the fly cannot eat unless the proboscis readout fires), it modulates steering and speed through the descending pools, and it decides whether a startle produces escape. What it does not do: fine motor coordination of walking, which is VNC-driven in real flies and kinematic rendering here.

2. Finding 1 — Global-max normalization at the reference gain does not propagate on aggregated connectomes

2.1 Measurement

The reference implementation normalizes edge weights globally: $w \leftarrow w / \max|w| \times 0.15$. On both aggregated datasets the raw synapse-count weights are extremely heavy-tailed:

dataset	max	median	p90	p99
FlyWire v783	2405	8	23	77–83 (sample-dependent)
MaleCNS v1.0	2591	9	28	93

Global-max normalization therefore maps the median edge to $\approx 0.15 \times 8/2405 \approx 5 \times 10^{-4}$ — roughly 1/2000 of the firing threshold (1.0). Empirically, driving the left olfactory pool at sustained intensity 0.5 for 100 ticks produced, on **FlyWire**, 18,525 spikes **all in sensory groups; central-brain spikes: 0**, with the downstream OLF_PN group's maximum membrane voltage observed within those 100 ticks reaching only **0.780** (0.688 within the first 50 ticks of the same run). On **MaleCNS** the same protocol produced 21,867 spikes, again all sensory, central and motor 0, with the 100-tick observed maximum at **0.977**. Signal dies in the antennal lobe at the reference gain on both datasets.

2.2 Postsynaptic L1 normalization (adopted in this project)

We replaced global scaling with **postsynaptic L1 normalization**: each postsynaptic neuron's total incoming $|w|$ is scaled to a constant `targetInput`. Attribution note: earlier versions of this note called this "Shiu et al. (2024)-style" — a misattribution. Shiu et al.'s published code (`philshiu/Drosophila_brain_model`, `model.py`) uses a fixed per-synapse coefficient ($w_{\text{syn}} = 0.275 \cdot \text{mV}$), not postsynaptic total-input normalization. The scheme adopted here is our own engineering choice for this project; the two differ fundamentally (per-neuron budget vs global coefficient). Sweeping `targetInput` (200-tick probes driving the same ORN pool):

targetInput	FlyWire (200 ticks)	MaleCNS (200 ticks)
1.0	MB_KC 2,544, GNG_DESC 0	MB_KC 148, DN 0
2.0	MB_KC 57,446, GNG_DESC 10	MB_KC 37,237, DN 0
3.0	GNG_DESC 3,810; stable over 500 ticks of maximal bilateral input	DN 11; no visual ignition
4.0	whole-brain ignition (VIS_ME 405,788 spikes/200 ticks with zero visual input)	DN 75; VIS_OL 2 (no ignition)

The working point is dataset-dependent: we ship 3.0 for FlyWire and 4.0 for MaleCNS (§5.3).

2.3 Gain dependence (review replication)

An external reviewer noted that the failure claim depends on the gain constant. Repeating the §2.1 protocol with the global-max coefficient scaled $\times 10$ ($0.15 \rightarrow 1.5$): **a small number of central spikes appears on both datasets — 16 on FlyWire, 131 on MaleCNS (plus 13 motor)** (node `game/tools/probe_gain.mjs`; archive `docs/results/probe_gain.txt`). We state this narrowly: at reference gain 0.15, propagation fails on both datasets; at $\times 10$, a small number of central spikes appears, and the stability or usefulness of the higher-gain regime is unknown (not characterized here). Accordingly, Finding 1 is scoped to the reference gain throughout.

2.4 Discussion

Aggregation is the culprit multiplier at the reference gain: collapsing all synaptic contacts between a neuron pair into one scalar produces a few huge weights (a 2405-contact edge) that set the global scale for everyone else, and a threshold model then divides the world into "edges that matter" and "edges that don't" with nothing in between. Any demo that "just runs dynamics" on an aggregated connectome should publish its normalization, its gain, and a propagation test; a silent network at reference gain is the default outcome, not a bug in your stimulus.

3. Finding 2 — Structural bias in coarse descending readouts (FlyWire measurement)

3.1 Measurement

Our FlyWire steering readout splits GNG_DESC (3,581 neurons) into left/right pools by soma x-coordinate median (1,791 / 1,790 neurons). Stimulating either antennal pool alone ($I=1.0$, 200 ticks, targetInput 3.0) gives:

stimulus	desc_left	desc_right	L/(L+R)
left antenna pool	1,704	2,106	0.447
right antenna pool	1,633	2,133	0.434

The right pool fires more under **either** unilateral stimulus (bias $\approx +24\%$ over the left pool), while the stimulus-dependent contrast between the two rows is only $\sim 1.3\%$. First descending spikes appear 9 ticks (~ 0.9 s at 10 Hz) after stimulus onset. Feeding the raw rates into a steering law $\text{turn} = k \cdot (R-L)/(R+L+\epsilon)$ would make the fly veer right forever regardless of where the banana is.

3.2 Heuristic high-pass readout

We steer on a **heuristic high-pass readout adopted in this project**: $\text{turn} = k \cdot (R^{\wedge} - L^{\wedge}) / (R^{\wedge} + L^{\wedge} + \epsilon)$ with $R^{\wedge} = \max(0, R - \text{EMA}_{\tau}(R))$, $\tau = 5$ s. We make no claim that this removes the bias or recovers a "true" lateral signal; it is a heuristic chosen for responsiveness, and the only ablation we offer is behavioral: longer baselines ($\tau = 20\text{--}40$ s) perform worse in our sweeps (eat rates fall to 40–80% in 7 of 8 swept configurations, and at $\tau = 40$ s path tortuosity rises to $\sim 17\text{--}23$), which we interpret as persistent wrong-sign deviations getting "locked in" by the long memory. Whether any baseline-free readout can recover steering contrast on this dataset is an open experiment (§6).

3.3 Interpretation — hypothesis, not conclusion

Where does the bias come from? Candidate contributors: the fly's largely bilateral olfactory pathway, our anatomically naive x-median split, the "GNG_DESC" group definition mixing descending and ascending neurons, and single-specimen hemispheric asymmetries in proofreading or annotation. v2 leaned toward "real structure"; v2's cross-dataset result (§5.4) instead suggests a pool-definition artifact. In v3 we explicitly downgrade all such attribution to **hypothesis**: the measurement itself stands, the explanation is undecided, and the decisive experiment is re-pooling/relabeling on the same FlyWire data (§6). Practical rule for demo builders, unchanged: **calibrate readout pools against a slow baseline and report the bias** — if your bilateral pools respond equally to both sides, your "steering" is cosmetic.

4. Finding 3 — Seed overfitting: a tuning-methodology lesson

4.1 The sweep winner that wasn't

Because steering is weak and fluctuation-dominated, time-to-first-banana is a high-variance metric (in our runs, 12–166 s across seeds for identical parameters; the game injects constant current, so the fluctuations arise from network dynamics and game-layer randomness, not from any Poisson implementation — v2's "Poisson noise" wording was imprecise). We swept game-layer readout constants ($\text{turnGain} \times \text{smellSigma} \times \text{baseSpeed}$, then $\text{turnGain} \times \text{emaTau} \times \text{baseSpeed}$; 12 configs $\times$ 5 LCG-seeded runs $\times$ 180 simulated seconds each, brain reset between episodes, full tables in `game/README.md`). The qualitative trends are modest and configuration-dependent: in our tables, higher `turnGain` is associated with worse outcomes in most rows, but there are counterexamples — e.g. at fixed `smellSigma=250` and `baseSpeed=12`, raising `turnGain` from 2.6 to 4 improves mean score from 1.80 to 2.20 — and enlarging `emaTau` degrades performance (§3.2). All first-eat statistics below count **successful episodes only** (a run with no eat contributes to the eat-rate but not to first-eat time). The apparent champion, `turnGain = 1.8`, scored 100% eat rate, 26.4 s median, 3.00 mean score across the 5 sweep seeds.

On 10 fresh seeds it collapsed: 8/10 ate, median 57.7 s, mean 73.7 s, mean score 1.30 — **worse than the status-quo default on the same 10 seeds** (10/10, 49.6 s median, 2.70 score). The 5-seed "win" was seed luck. A configuration-reproducibility caveat adds a second layer: the champion's 3.00 was measured when `bananaMaxDist` was still unbounded; under the current default (`bananaMaxDist = 250`) the same configuration scores 2.20, and explicitly restoring the historical unbounded value recovers 3.00 (both numbers match the reviewer's independent measurements; commands in §7).

4.2 What actually helped: level design

Capping banana spawn distance at 350 px excluded the far corners of the arena (max distance from the center-spawned fly is ≈ 362 px), yet produced metrics identical to the default in our seed set — no measurable effect. Capping spawns to a 150–250 px ring, however, moved the median time-to-first-banana from 49.6 s to **27.3 s** (10/10 eaten, tortuosity 8.47 — still clearly wandering search rather than a ≈ 1 straight-line beeline). The "fast search" family (higher speed gain / max speed, rewriting the rate $\rightarrow$ speed readout mapping) tells a mixed story and was rejected: fast-search alone improved the median (38.6 s < 49.6 s) but reduced the eat rate to 9/10 and the mean score to 1.80; ring+fast combined was worse than ring-only on the median (42.6 s > 27.3 s) while scoring 3.10 — a score gain we attribute to faster wandering covering the ring sooner, not to any readout improvement, and we did not adopt it.

4.3 Honest-tuning principle

The final shipped defaults change only the level-design constant (`bananaMaxDist = 250`). Every constant in the sensory $\rightarrow$ brain $\rightarrow$ readout mapping is unchanged, so the demo's core claim — "the brain's readout participates in the decision" — carries no hidden tuning water. Two methodological lessons generalize beyond games. First: **behavioral metrics with high stochastic variance must be validated out-of-sample; a small-seed sweep does not estimate the parameter effect, it estimates the seed set**. Second, disclosed in v3: **the seeds (101–110) used to select the level-design constant were also used to evaluate it — they are no longer an independent test set**. We kept both tables in the repo README precisely so the selection is auditable.

5. Cross-dataset replication on MaleCNS

Everything in §2–§4 was first measured on FlyWire. We then re-ran the same probe battery on MaleCNS v1.0 through the identical simulation core. Full tables and commands are in `docs/malecns-probes.md` and `docs/results/`; the highlights and their consequences follow. §5.1–§5.4 were substantially rewritten in v3 after external review.

5.1 Preprocessing caveats (corrected in v3)

Both datasets are ≥ 5 -synapse filtered. v2 claimed the FlyWire package "keeps aggregated connections of any strength" — that was wrong. Unsigned aggregation of the FlyWire source CSVs yields a minimum connection weight of 5.0 with zero neuron pairs below 5 (2,700,513 unsigned pairs; the 2,277-pair difference to the binary's 2,698,236 edges is pairs whose signed weights cancel to exactly zero and are dropped). The MaleCNS bundle is likewise thresholded at ≥ 5 synapses (6,287,749 connections) and additionally drops autapses.

The neurotransmitter sign rules differ between packages — a major confound.

transmitter	FlyWire package (snedea NT_SIGN)	MaleCNS package (fly-brain-minecraft)
acetylcholine	+1 excitatory	+1 excitatory
glutamate	+1 excitatory	-1 inhibitory
histamine	not a predicted class in this dataset	-1 inhibitory (8,021 labeled nodes)
GABA	-1 inhibitory	-1 inhibitory
monoamines (DA/OA/5-HT)	+1 excitatory	+1 excitatory

Correction history on histamine (kept for transparency): v3 incorrectly claimed both packages treat histamine as inhibitory; v4 over-corrected by adding "no entry → default +1, photoreceptors directly affected". Both were wrong in different ways. The facts, verified by scanning every `nt_type` label in the FlyWire source `connections.csv.gz` (3,869,878 rows): the label set is exactly six classes — ACH 2,258,155; GABA 865,318; GLUT 654,183; DA 37,705; SER 37,450; OCT 17,067 — **with no histamine and no empty labels**. FlyWire's six-class transmitter prediction simply does not include histamine, and neurons (including photoreceptors) are labeled with one of these six classes; **the absence of a predicted class cannot be used to infer how real histaminergic neurons are signed in the FlyWire package**. The MaleCNS package, by contrast, explicitly assigns -1 to 8,021 histamine-labeled nodes (that count is the total of histamine-labeled nodes, not a photoreceptor count). Glutamatergic neurons are abundant in both datasets (MaleCNS: 29,763 neurons, ~17%). Cross-dataset dynamical differences below must be read against these sign-table differences. Other differences remain: sex (female/male), VNC (absent/present), individuals, and packaging pipelines.

5.2 Finding 1 — replicated at reference gain, and gain-bounded

Covered in §2.1 and §2.3: global-max normalization fails at the reference gain on both datasets (100-tick observed maxima 0.780 / 0.977, zero central spikes), and a small number of central spikes appears at 10× gain (16 / 131 central spikes). The cross-dataset statement that survives review is therefore narrow but robust: **at the reference gain of the reference implementation, global-max normalization does not propagate on either aggregated connectome**.

5.3 targetInput — structure replicates, the working point drifts; sign rules, not filtering, drive the ignition difference

The per-neuron sweep (§2.2 table) reproduces the same layered reachability on MaleCNS — 1.0 dies at the mushroom-body entrance, 2.0 ignites the mushroom body but not descending neurons, 3.0 first reaches them (DN 11), 4.0 makes them usable (DN 75). Two quantitative differences from FlyWire:

First, the olfactory→descending-neuron drive is **about two orders of magnitude weaker** (11 vs 3,810 GNG_DESC spikes per 200 ticks at $t_i=3.0$). Candidate contributors: MaleCNS's `descending_neuron` superclass is a pure brain→VNC command population several synapses from the olfactory entry, whereas FlyWire's "GNG_DESC" group mixes in directly sensory-driven neurons; and the sign-rule difference changes net excitation everywhere.

Second — and here v2's attribution must be retracted — **MaleCNS shows no FlyWire-style high visual-lobe activity at $t_i=4.0$** (VIS_OL: 2 spikes, versus VIS_ME 405,788 spikes/200 ticks on FlyWire). v2 attributed this to MaleCNS's ≥ 5 -synapse filtering; that attribution was void (§5.1: both datasets are filtered). The replacement hypothesis was the sign-table difference, and we tested it in both directions:

Forward experiment (sufficiency, MaleCNS). We regenerated the MaleCNS binary with only the glutamate sign flipped from -1 to +1 (`python game/tools/flyb_to_bin.py --glut-excitatory`, 29,763 neurons flipped; all other parameters identical), and ran the $t_i=3/4$ probes:

variant	total spikes (200 ticks)	DN	VIS_OL
MaleCNS standard (GLUT inhibitory), $t_i=3.0$	235,638	11	0
MaleCNS standard, $t_i=4.0$	331,336	75	2
MaleCNS GLUT-excitatory, $t_i=3.0$	492,145	1,025	41
MaleCNS GLUT-excitatory, $t_i=4.0$	1,797,262	19,004	553,863

Reverse experiment (necessity, FlyWire, added in v4). We rebuilt the FlyWire binary from the source connection table with only the glutamate sign flipped from +1 to -1 — i.e. the source rows are re-aggregated and postsynaptic-L1-normalized identically, differing only in the sign rule (`python`

game/tools/build_flywire_variant.py; equivalence self-check: a standard-sign rebuild reproduces the shipped connectome.bin.gz edge-for-edge, 2,698,236 edges, zero mismatches), and ran the same probes:

variant	total spikes (200 ticks)	GNG_DESC	VIS_ME
FlyWire standard (GLUT excitatory), ti=3.0	447,219	3,810	2,860
FlyWire standard, ti=4.0	1,265,998	25,730	405,788
FlyWire GLUT-inhibitory, ti=3.0	222,036	30	0
FlyWire GLUT-inhibitory, ti=4.0	291,831	113	0

Scope note. The intervention is a single rule change, not a single-edge change: it alters the edge set by 517 edges (2,698,753 vs 2,698,236, from changed cancellation), the weights of 77,650 shared edges, and 23,905 postsynaptic L1 denominators. These are consequences of the same rule intervention, not an additional confound — but "the graph" here therefore means the fixed source connection table re-aggregated under each sign rule, not one frozen weighted graph.

What these two experiments establish, stated as narrowly as the data allow: **at ti=4, l=1, 200 ticks, and across the two GLUT sign configurations compared, flipping only the glutamate sign is sufficient to cause high visual-lobe activity on the MaleCNS-derived graph, and flipping it back abolishes that activity on the FlyWire-derived graph (405,788 → 0) — the glutamate sign assignment is both sufficient and necessary for the high-activity regime under exactly these conditions.** The ti=3 rows are reported for completeness and are not part of this conclusion. This directly supports sign assignment as a major contributing factor to the cross-package dynamical difference; it does not prove sign tables explain all cross-dataset differences (drive asymmetry, anatomy, and filtering still differ). Excitability is partly a packaging choice: Shiu et al. assign glutamate to the inhibitory category (GluCI), the FlyWire browser package to the excitatory one, and the resulting networks sit in measurably different dynamical regimes. (node game/tools/probe_signflip.mjs; archives docs/results/probe_signflip.txt, docs/results/signflip_flywire.txt.)

The practical advice to the community sharpens accordingly: **run a per-dataset propagation test, publish the working point and the sign table — there is no universal constant, and there is no universal sign convention either.**

5.4 Finding 2 — not replicated under the tested MaleCNS conditions

With MaleCNS's native somaSide pool split and adequate drive (ti=4.0), the lateral contrast is **correctly signed**:

drive	stimulus	desc_left	desc_right	L/(L+R)
ti=3.0, 200 ticks	left antenna pool	5	6	0.455
ti=3.0, 200 ticks	right antenna pool	12	15	0.444
ti=4.0, 400 ticks	left antenna pool	91	66	0.580
ti=4.0, 400 ticks	right antenna pool	111	118	0.485

The left–right difference between the two ti=4 rows is $91/157 - 111/229 \approx 0.09490$ (**9.49 percentage points**). At ti=3.0 the DN counts are too small (5–15 spikes per pool in 200 ticks; 10–34 in 400 ticks) for any bias statistic.

What can and cannot be concluded. Under the tested MaleCNS conditions, the wrong-direction bias does not appear. But the two measurements differ in dataset, individual, sex, VNC presence, **sign tables**, pool definitions, and working point all at once — so "pool-definition artifact" (v2's preferred reading) remains a **hypothesis, not a conclusion**. The decisive experiment changes only the pooling on a fixed dataset: re-split FlyWire's descending population by alternative rules (e.g. hemispheric relabeling, ascending-excluded re-pooling) and re-measure (§6, next experiments). One related observation, kept strictly within MaleCNS: its ascending neurons (AN group) fired 0 times in 200 ticks under olfactory stimulation (versus 710 under bristle stimulation). Whether FlyWire's mixed GNG_DESC group derives any of its olfactory responsiveness from its own ascending component **has not been verified** — we removed the v3 sentence that claimed otherwise.

Statistical disclosures. (i) The MaleCNS olfactory stimulus pools are asymmetric: 411 unknown-side ORNs were assigned to the left/right pools by index parity, giving 1,090 (left) vs 1,549 (right) neurons — at equal per-neuron current, the right pool receives $\approx 42\%$ more total input. (ii) 10 midline descending neurons were assigned to the pools by parity. (iii) Each probe condition is a single deterministic trajectory; spike counts

within a run are not independent samples, and $t_i=4.0$ does not by itself make the comparison statistically adequate. (iv) The heuristic high-pass readout of §3.2 remains the readout used in the game on both datasets.

5.5 Biological sanity checks (small, but free)

- **DNp01 (giant fiber) does not respond to bristle stimulation:** 0 spikes in 20 ticks of strong mechanosensory-bristle drive — consistent with its known biology as a visually (looming-) triggered escape neuron. The game's escape path therefore keys on the descending-rate surge (bristle drives ascending/central activity strongly, 710 AN spikes in the same probe), not on DNp01.
- **The feeding chain works on MaleCNS:** gustatory pool at $I=1.2$ drives the MN9-class proboscis motor readout (16 cb_motor neurons) to 3 spikes in 30 ticks (first at tick 22) — at the game's feeding-gate threshold, and the headless closed loop on MaleCNS eats bananas for real (score 1–3 per 180 s across seeds, first eats 49–172 s; steering is legitimately "hard mode": DN rates of ~2–8 spikes/s at distance, rising to 34–42/s at close range).

6. Limitations and open questions

Limitations. Point LIF neurons without conductances, delays, neuromodulation, or plasticity; no gap junctions; transmitter signs from prediction pipelines with package-specific sign rules that materially change dynamics (§5.1, §5.3); two individuals (one female FAFB, one male MaleCNS) with different packaging pipelines; FlyWire has no ventral nerve cord, and even on MaleCNS we do not simulate real gait; **the game contains explicit motor rules (§1.1)**; per-dataset stimulus gains are acknowledged level design; probe conditions are single deterministic trajectories, and the level-design seed set doubles as its evaluation set (§4.3). None of this claims biological fidelity, let alone "uploading".

Next experiments (ordered):

1. **Decisive test for the Finding 2 attribution:** on the same FlyWire data, re-pool the descending population under alternative rules (hemispheric relabeling, ascending-excluded, native-side annotations from FlyWire's own classification) and re-measure the unilateral-stimulus contrast. If the wrong-direction bias vanishes under alternative pooling, the artifact reading is confirmed on that dataset.
2. **BANC replication** of the global-max reference-gain failure (one-command dataset swap), plus a record of each packaging's sign table as a mandatory companion to any dynamical comparison.
3. **Named-DN readouts:** which specific descending types (e.g. the turn/halt DNs characterized in the literature) restore steering contrast without any baseline subtraction, on either dataset?
4. **Gain stability:** characterize whether the $10\times$ global-max gain (§2.3) yields usable, non-seizing dynamics or merely pushes the failure threshold higher.

7. Reproducibility

All numbers in this note were produced on this repository (Node v24, Python 3.13, Windows; CPU-only; minutes), **except where a row is explicitly marked as a historical record**. Commit lineage (v1 → this revision) is given in the header. Vendor data acquisition with pinned commits and SHA-256 checksums: `game/tools/fetch_vendor_data.md`. Probe batteries are frozen as scripts (`probe_flywire.mjs`, `probe_malecns.mjs`, `probe_gain.mjs`, `probe_signflip.mjs`, `probe_supplementary.mjs`, `reproduce_sweeps.mjs`); key per-run outputs are archived under `game/docs/results/` with the producing command in each file's header. Timing fields (ms/tick) fluctuate with machine load and are not bit-stable; all spike counts are.

number(s)	command
FlyWire propagation (47,106 spikes / 35,159 central); N/E counts	<code>node game/test/sim.test.mjs</code>
FlyWire weight distribution; reference-gain failure; targetInput sweep; bias table	<code>node game/tools/probe_flywire.mjs</code> (archive: <code>docs/results/probe_flywire.txt</code>)
gain $\times 1/\times 10$ retest, both datasets (16 / 131+13 central)	<code>node game/tools/probe_gain.mjs</code> (archive: <code>docs/results/probe_gain.txt</code>)
FlyWire pools generation (weight scan incl. p99)	<code>python game/tools/prepare_pools.py</code>
MaleCNS three-file conversion (176,422 / 6,287,749; weight stats)	<code>python game/tools/flyb_to_bin.py</code>
MaleCNS probes 1–4 (distribution; reference-gain failure; targetInput; bias)	<code>node game/tools/probe_malecns.mjs</code> (archive: <code>docs/results/probe_malecns.txt</code>)

number(s)	command
sign-flip variants: MaleCNS GLUT-excitatory build	<code>python game/tools/flyb_to_bin.py --glut-excitatory</code>
FlyWire GLUT-inhibitory rebuild + equivalence self-check (2,698,236 edges, 0 mismatches; variant +517 edges)	<code>python game/tools/build_flywire_variant.py</code>
sign-flip bidirectional probe (both directions, $t_i=3/4$)	<code>node game/tools/probe_signflip.mjs</code> (archives: docs/results/probe_signflip.txt, docs/results/signflip_flywire.txt)
MaleCNS feeding/escape/ascending supplementary probes (feed 3/tick 22; DNp01 0, AN 710; AN 0, DN 75)	<code>node game/tools/probe_supplementary.mjs</code> (archive: docs/results/probe_supplementary.txt)
FlyWire closed loop (seed 7: first eat 34.8 s, score 2/120 s)	<code>node game/tools/headless_run.mjs 120 '{}' 7 flywire</code>
MaleCNS closed loop (score 1, first eat 104.3 s)	<code>node game/tools/headless_run.mjs 180 '{}' 7 malecns</code>
sweep round 1 (turnGain $\times$ smellSigma $\times$ baseSpeed)	<code>node game/tools/tune_sweep.mjs '{"turnGain":[2.6,4,6],"smellSigma":[130,250],"baseSpeed":[6,12],"bananaMaxDist":[null]}'</code>
sweep round 2 (turnGain $\times$ emaTau $\times$ baseSpeed)	<code>node game/tools/tune_sweep.mjs '{"turnGain":[1.8,2.6],"emaTau":[5,20,40],"baseSpeed":[6,12],"bananaMaxDist":[null]}'</code>
champion config, historical default restored (mean score 3.00; under current default 250 it is 2.20)	<code>node game/tools/tune_sweep.mjs '{"turnGain":[1.8],"emaTau":[5],"baseSpeed":[6],"bananaMaxDist":[null]}'</code>
out-of-sample & level-design comparisons (bananaMaxDist $\in \{\infty \rightarrow \text{null}, 350, 250\}$; 10 seeds $\times$ 180 s; reproduces the §4.2 table exactly)	<code>node game/tools/reproduce_sweeps.mjs</code> (archive: docs/results/reproduce_sweeps.txt)

Notes on exactness. "bananaMaxDist": [null] in the sweep commands explicitly restores the historical unbounded spawn distance ($\text{null} \rightarrow \infty$ in harness.mjs); without it, today's default (250) silently changes the historical tables — the reviewer's 2.20-vs-3.00 discrepancy was exactly this. Seeded runs use an injectable LCG replacing Math.random; episodes call sim.reset() between runs so one process reuses a single parsed brain. Headless runs are not real-time throttled. Honest reuse note: seeds 101–110 served both for selecting bananaMaxDist and for evaluating it (§4.3); first-eat statistics count successful episodes only. One row is a **historical record not rebuilt by script**: the 350 px no-effect comparison of §4.2 (its configuration is documented in the game/README.md tuning log); every other number in the tables above is reproduced by the listed commands today.

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Community context: this note was written against the backdrop of the 2026 "flyslop" demo wave (indexed at awesome-fly and Fly Brain Hub, <https://fly-brain-hub.vercel.app/>) and community validation efforts such as flybench. v3 and v4 were revised in response to two rounds of external methodological review; the point-by-point response letter is `docs/REVIEW-RESPONSE.md`.